Business Requirements Document

NLNG Experience Center — Internal Audit Continuous Monitoring Simulation

Scenario 9: Ghost Worker & Payroll Anomaly Monitoring

Client	Nigeria LNG (NLNG)
Engagement	NLNG Internal Audit Experience Center — Demonstration Build
Process Area	Payroll / Workforce Management (supporting scenario, outside core Contract Management / Asset & Maintenance Management scope)
Prepared For	Development Team
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Status	Draft for Review

1. Purpose and Background

1.1 Purpose

This document defines the business requirements for simulating Scenario 9 — Ghost Worker & Payroll Anomaly Monitoring — within the NLNG Internal Audit Experience Center. It is intended to give the development team a complete, unambiguous specification covering the source-system simulation, the audit monitoring tool behavior, detection logic, and the end-to-end demonstration flow, so the correct solution can be built without further clarification.

1.2 Background

NLNG has requested an Experience Center demonstration focused on building internal audit capability through continuous, real-time monitoring rather than periodic sampling. The core engagement scope covers Contract Management and Asset & Maintenance Management. In addition, three supporting scenarios outside these two areas were agreed to round out the demonstration and show the breadth of the audit monitoring tool across different business processes. Scenario 9 (Payroll/Workforce) is one of these three supporting scenarios.

NLNG has explicitly requested an interactive, hands-on experience rather than a slide presentation. The audience should be able to see a process being initiated, observe where it deviates from policy, and watch the audit monitoring tool detect, investigate, and track the

issue — while preserving the principle that audit raises and advises on exceptions but does not resolve them directly, to maintain independence from the business process.

1.3 Objective of This Scenario

To simulate a believable hire-to-payroll process and demonstrate how a continuous monitoring tool detects two specific anomalies that a manual, sample-based review would likely miss: (1) ghost workers — employees who remain active and paid in payroll despite exit/termination, and (2) duplicate or shared bank accounts across supposedly distinct employee records.

2. Scope

2.1 In Scope

- A mock HR/Payroll (HRIS) application simulating new hire creation, hire approval, exit/termination processing, and the monthly payroll run.
- The Audit Monitoring Tool's ingestion, detection, alerting, drill-down, and case-tracking functionality for this scenario.
- Two anomaly types: Ghost Worker (active payroll record with no corresponding current employment) and Duplicate/Shared Bank Account.
- Pre-loaded / seeded sample data sufficient to run the demonstration live, plus the ability to trigger a live anomaly during the session.

2.2 Out of Scope

- Integration with any real NLNG system, including SAP, actual HRIS, or actual payroll/banking platforms.
- Actual salary payment processing, GL posting, or financial settlement.
- Automated resolution or blocking of transactions by the audit tool (the tool must only alert and track, never act unilaterally — see Section 9.2).
- Scenarios 1–8 and Scenario 10 (Contract Management, Asset & Maintenance Management, and other supporting scenarios) — covered under separate BRDs.

3. Personas and Roles

The following personas must be represented, either as distinct UI roles/logins or as clearly distinguishable actors within the demo narration.

Persona	Description & System Role
HR Officer (Process Owner — Hire/Exit)	Creates new hire records and processes exits in the HRIS app. Represents the process point where ghost workers originate, whether through administrative lag or deliberate manipulation.
Second HR Approver (Verification Authority)	Reviews and signs off on new hire records before they go live in payroll, in the HRIS app. High hire volume across a large contractor workforce means verification is brisk, not forensic.

Payroll Processor	Runs the monthly payroll cycle in the HRIS app against all active employee records, trusting HR's active/inactive status is accurate. Not shown performing detection — included for process completeness only.
Terminated/Fictitious Worker (Risk Subject)	Not a system user — exists only as a record. Represents either a genuine exit not yet processed, or a fabricated identity used to siphon payroll funds.
Auditor (Continuous Monitoring Reviewer)	Primary user of the Audit Monitoring Tool. Reviews system-raised exceptions, drills into record detail, classifies root cause, scores risk, and refers findings — without intervening in the business process.
Business Owner (HR/Payroll Lead — Referral Recipient)	Receives the auditor's referral in the Audit Monitoring Tool and is responsible for investigating and actioning the finding (deactivating records, recovering funds). Owns resolution, not the audit team.
Audit Monitoring Tool	System persona. Continuously cross-references HR and payroll records, runs detection rules, raises exceptions, and tracks case status end-to-end. Never auto-resolves or blocks a payroll transaction.

3.1 Persona Narratives

Fuller descriptions of each persona's role within the demo story, for use in presenter scripting and character briefing:

1. The HR Officer (Process Owner — Hire/Exit)

Mr. Chukwudi represents HR staff who create new hire records and process exits. Diligent in the clean scenario, but represents the process point where ghost workers originate — whether through genuine administrative lag (delayed exit paperwork) or deliberate manipulation of hire/exit records.

2. The Second HR Approver (Verification Authority)

Reviews and signs off on new hire records before they go live in payroll. Handles many hires across a large, high-turnover contractor workforce, so verification is necessarily brisk. This persona is structurally unlikely to catch a duplicate bank account unless it's cross-checked against the full employee file.

3. The Payroll Processor

Runs the monthly payroll cycle against all “active” employee records, trusting that HR's active/inactive status is accurate and current. Operates downstream of HR, with no independent means of verifying whether an “active” employee is still genuinely on site or real — making this the point where ghost payments actually get released.

4. The Terminated/Fictitious Worker (Risk Subject)

Represents either a genuine contractor whose exit wasn't processed in time, or a fabricated identity created to siphon payroll funds. Not an active system user — this persona exists only as a record in the system, which is precisely what makes the anomaly hard to catch through normal observation.

5. The Auditor (Continuous Monitoring Reviewer)

Reviews exceptions the tool raises across the full payroll population, not a sample. Investigates flagged cases, classifies root cause, scores risk, and refers findings to HR/Payroll — while not intervening directly in hiring, payroll processing, or account verification, to preserve independence from the business process.

6. The Business Owner (HR/Payroll Lead — Referral Recipient)

The HR or Payroll lead who receives the auditor's referral and is responsible for investigating — confirming whether a flagged case is a delayed exit, a system gap, or deliberate fraud. This persona owns the actual correction (deactivating records, recovering funds), not the audit team.

7. The Audit Monitoring Tool (System Persona)

The “silent observer” cross-referencing HR exit records, payroll activity, and bank account details across the entire workforce continuously. Matches patterns humans would miss — shared accounts, stale “active” records — flags exceptions instantly, and tracks each case's status end-to-end, without ever acting unilaterally on payroll itself.

4. Narrative Walkthrough (Demo Story)

This section translates Sections 5-8 into the story the presenter will actually narrate on the day. It is intended to help the development team understand the intended emotional/dramatic arc of the demo, so screens, pacing, and transitions are built to support it — not just the underlying logic.

Act 1: The Process, As It Should Work

Picture Mr. Chukwudi, an HR officer at NLNG's Bonny Island site. A new technician, Emeka, joins the maintenance contractor team.

1. Initiation — Chukwudi raises a new hire record in the HR/payroll system: name, role, department, bank account, start date, and reporting manager — all tied to an approved headcount requisition.
2. Verification — A second HR approver reviews and signs off on the new hire record before it becomes “active” in payroll.
3. Payroll run — Each month, payroll pulls all active employee records and processes salary payments against each employee's verified bank account.
4. Exit handling — When a contractor's assignment ends, HR deactivates the record, and payroll excludes them from the next run.
5. Close — Nothing unusual. Every active record maps to a real, currently working person, and every payment goes to a verified, unique account. This is the baseline the audit tool compares every payroll run against.

Act 2: Where It Goes Wrong

Two realistic ways this breaks down:

Risk 1 — The Ghost Worker.

A contractor's assignment ends, but the exit paperwork is delayed, skipped, or simply not processed in payroll. The record stays “active,” and salary keeps being paid into the same account every month — for someone no longer on site. In a large workforce with constant contractor turnover, a handful of these can persist for months unnoticed.

Risk 2 — The Duplicate/Shared Account.

Two “different” employee records — perhaps one real, one fabricated, or two legitimate-looking entries created by the same person with access — are both paid into the same bank account. On paper, payroll sees two distinct employees; in reality, one person (or a colluding pair) is collecting both salaries.

Neither looks wrong on a payroll report. The ghost worker's payment looks identical to every other legitimate payment. The duplicate account only becomes visible if someone deliberately cross-references account numbers across the entire employee file — something a monthly manual payroll review, done by sampling, is very unlikely to catch.

Act 3: The Audit Tool Picks It Up

The moment the payroll run is ingested, the audit monitoring tool runs its checks across the entire population, not a sample:

- For the ghost worker risk: it cross-references active payroll records against HR exit/termination records, attendance or site-access logs where available, and time since last activity — flagging any employee paid with no corresponding recent activity or an exit record that was never reflected in payroll.
- For the duplicate account risk: it matches bank account number (and other identifiers like tax ID or next-of-kin details) across all employee records — flagging any account tied to more than one “different” employee.

Both exceptions appear on the dashboard the moment that payroll run is processed — not at year-end audit.

Act 4: What the Audit Tool Actually Does Next

This is the part NLNG specifically cares about, and it's worth being precise here, because the tool must notify, not resolve — auditors stay independent and advise the business owner rather than stepping into the process themselves. The response sequence looks like this:

6. Alert, not auto-action. The tool doesn't stop the payroll run or suspend the payment. It raises a prompt alert to the audit queue.
7. Drill-down. The auditor opens the case and sees the two flagged records side-by-side — shared bank account highlighted, or the mismatch between “active” payroll status and “terminated” HR status clearly shown, with a direct link to both source records.
8. Root cause classification. The tool helps classify the case: administrative lag (genuine error — exit paperwork simply delayed), process/system gap (HR and payroll systems not properly linked), or probable deliberate fraud (shared account, fabricated identity details).
9. Risk scoring. Scored by cumulative amount already paid, duration the anomaly has existed, and whether this is a one-off or a recurring pattern for this department/approver.

10. Referral, not resolution. The case is referred to the relevant HR/Payroll business owner with full evidence — the audit team advises, it doesn't suspend payroll itself.
11. Track to resolution. Status moves through Flagged → Under Review → Referred → Response Received → Closed/Escalated, so a ghost worker case can't quietly disappear into an inbox.

That's the full loop: an invisible-to-manual-review payroll leak, caught the moment the anomaly exists, investigated with full traceability, and handed to HR/Payroll to act on — while audit stays the advisor, not the enforcer.

Presenter Pacing Guidance

Act 1 should be fast and light — narrated in under 30 seconds with at most 1-2 screens, since the demo's value lies in Acts 2-4, not in watching a routine hire get processed. The bulk of live interaction time should be spent on the deviation occurring, the tool catching it, drill-down, alert, and resolution tracking.

5. Systems to Be Simulated

Two systems/applications must be built for this scenario:

12. Mock HR/Payroll (HRIS) Application — stands in for NLNG's real HR and payroll system.
13. Audit Monitoring Tool — the central tool shared across all demonstration scenarios; this BRD defines only the Payroll/Workforce detection logic and workflows within it.

Note: the Audit Monitoring Tool is a shared, central system across all 10 scenarios. Its core case-management, dashboard, and alerting framework should be built once and configured per scenario — not rebuilt for Scenario 9 specifically. This BRD specifies the configuration and detection logic this scenario requires of that shared tool.

6. Process Flow — Normal (Baseline) Path

This is the control path. It must be demonstrable quickly (verbally narrated, with at most 1–2 screens shown) since the demonstration's value lies in the deviation and detection, not the routine flow.

Step	Actor	Action
1	HR Officer	Raises a new hire record in the HRIS app: name, role, department, bank account, start date, reporting manager, linked to an approved headcount requisition.
2	Second HR Approver	Reviews and approves the new hire record; it becomes active in payroll.
3	Payroll Processor	Monthly payroll run processes salary payment against the employee's verified bank account.

4	HR Officer	When the contractor's assignment ends, HR Officer deactivates the employee record.
5	Payroll Processor	Next payroll run excludes the deactivated record. Close — no exception generated.

7. Process Flow — Anomaly Paths

7.1 Anomaly 1: Ghost Worker (Active Payroll Record Post-Exit)

Business rule context: every employee record in payroll must correspond to a currently employed individual; HR exit processing should deactivate payroll eligibility immediately upon assignment end.

Step	Actor	Action
1	Terminated Worker	Contractor assignment ends on site.
2	HR Officer	Exit paperwork is delayed, skipped, or not processed in the HRIS app; the employee record remains “active.”
3	Payroll Processor	Monthly payroll run includes the still-active record and processes payment as normal.
4	Audit Monitoring Tool	On ingestion of the payroll run, cross-references active payroll records against HR exit/termination records and activity indicators; detects an active, paid record with no corresponding current employment. Raises exception (see Section 8.1).

7.2 Anomaly 2: Duplicate / Shared Bank Account

Step	Actor	Action
1	HR Officer	Creates a new hire record (genuine or fabricated) with a bank account that is already associated with another active employee record.
2	Second HR Approver	Approves the record — the shared account is not visible without cross-referencing the entire employee file.
3	Payroll Processor	Processes payment for both records into the same account.
4	Audit Monitoring Tool	Matches bank account number (and other identifiers, e.g. tax ID) across all employee records; detects the shared account across two distinct employee IDs. Raises exception (see Section 8.2).

8. Detection Logic / Business Rules

8.1 Rule: Ghost Worker Detection

- Trigger: a payroll run is ingested by the Audit Monitoring Tool.
- Matching key: Employee ID, cross-referenced against HR exit/termination records, site-access or attendance logs (where available), and last recorded activity date.
- Logic: for each active, paid employee record, check for a corresponding exit/termination record, or for an elapsed period since last recorded activity beyond a configurable tolerance.
- Exception condition: a payroll record is active and paid, but has a matching exit record not reflected in payroll status, or shows no activity for longer than the configured tolerance window.
- Severity: Critical (direct, ongoing financial loss for every payroll cycle the record remains active).

8.2 Rule: Duplicate / Shared Bank Account Detection

- Trigger: a new or amended employee record with bank account details is ingested, or a payroll run is processed.
- Matching key: Bank account number + bank name, cross-checked against all other active employee records; secondary check on tax ID / next-of-kin details where available.
- Logic: compare the incoming or existing account details against the full employee population, regardless of department or hire date.
- Exception condition: the same bank account is associated with more than one distinct, active employee ID.
- Severity: Critical (strong indicator of fraud or collusion).

8.3 Root Cause Classification (Tool-Assisted)

The tool should support the auditor in classifying each exception into one of the following categories, using pattern indicators as a guide (not an automated final determination):

Category	Pattern Indicators
Administrative Lag (Genuine Error)	Single, recent occurrence; exit record exists but was processed late; no history of similar cases for this HR officer/department.
Process/System Gap	Pattern occurs across multiple departments or HR officers, suggesting the control itself (e.g. no automated link between HR exit and payroll deactivation) is the root cause rather than individual behaviour.
Probable Deliberate Fraud	Shared bank account across “different” employees; exit record never created despite clear evidence of departure; repeat pattern linked to the same HR officer or approver.

8.4 Risk Scoring Factors

- Cumulative value already paid against the flagged record since the anomaly began.
- Duration — how many payroll cycles the anomaly has persisted undetected.
- Frequency — is this a first occurrence or a repeat pattern for this department, HR officer, or approver?

- Whether the pattern is isolated or occurs across multiple employees (systemic indicator).

9. Audit Monitoring Tool — Required Workflow

The following case lifecycle must be supported and visibly demonstrable on the Audit Monitoring Tool dashboard:

14. Flagged — exception is raised automatically per Section 8 logic; appears on dashboard in near real time.
15. Under Review — auditor opens the case; tool presents side-by-side comparison of the matched employee records with matching fields (e.g. shared bank account, conflicting status) highlighted, and a direct link back to the source records in the HRIS app.
16. Root Cause Classified — auditor assigns a category per Section 8.3.
17. Risk Scored — case is scored per Section 8.4 factors; priority assigned.
18. Referred — case and full evidence trail is sent to the relevant HR/Payroll Business Owner for investigation and response. The tool must not take any corrective action itself.
19. Response Received — Business Owner's input is logged against the case.
20. Closed / Escalated — case is closed with outcome notes, or escalated if unresolved/high-risk.

9.1 Dashboard Requirements

- Live exception feed showing newly flagged cases as they are generated.
- Trend view showing exception volume/type over time (to support the 'trend and anomaly detection' capability NLNG asked to see).
- Drill-down from any dashboard tile directly to record-level detail.
- Case status tracker visible per case (per the 7-stage lifecycle above).

9.2 Non-Negotiable Design Principle: Audit Independence

The Audit Monitoring Tool must only notify and track. It must never automatically deactivate a record, block a payment, or otherwise act on the business process. This preserves the auditor's role as independent advisor to the business owner, which NLNG explicitly required.

10. Sample Data Requirements

10.1 HR/Payroll (HRIS) App — Required Data Fields

Entity	Required Fields
Employee Record	Employee ID, Name, Department, Role, Bank Account Number, Bank Name, Tax ID, Start Date, Reporting Manager, Status (Active/Inactive), Approval Status, Approver ID

Exit / Termination Record	Exit ID, Linked Employee ID, Last Working Date, Exit Reason, Processed Date, Processed By
Payroll Run	Run ID, Period, Linked Employee ID, Amount Paid, Bank Account Used, Payment Date, Payment Status

10.2 Seed Data Scenarios to Pre-Load

- 10-15 normal, clean active employee records across multiple departments (baseline, no exceptions).
- 1-2 pre-seeded ghost worker examples — an active, paid record with a corresponding exit record not reflected in payroll (for fallback/safety if live demo entry fails).
- 1-2 pre-seeded duplicate bank account examples (same, as fallback).
- Ability for the presenter to enter a new live exit record or new hire record during the session and have the next payroll run evaluated against existing seed data in real time, to demonstrate true continuous monitoring.

11. Demonstration Flow (UX Sequencing Guidance)

This section guides screen/interaction design so the live walkthrough matches the intended narrative pacing.

21. Brief orientation (~10-15 seconds): narrate the normal process (Section 6) using at most 1-2 screens. Do not fully click through this path live.
22. Live trigger: presenter enters the anomaly (a ghost worker scenario via an unprocessed exit, or a duplicate bank account on a new hire) in the HRIS app.
23. Immediate detection: switch to the Audit Monitoring Tool dashboard; the new exception should appear with minimal delay once the payroll run is processed.
24. Drill-down: open the case; show the matched employee records side-by-side with matched fields highlighted.
25. Root cause & risk scoring: show classification and scoring applied to the case.
26. Referral: show the case being referred to the HR/Payroll Business Owner persona, with the tool explicitly not auto-resolving.
27. Resolution tracking: show the case status progressing (e.g. to 'Response Received' or 'Closed'), demonstrating full traceability.

12. Acceptance Criteria

- A ghost worker scenario entered live in the HRIS app is correctly detected and flagged by the Audit Monitoring Tool within the demo session, with the conflicting active/exit status clearly shown.
- A duplicate bank account entered live is correctly matched against an existing employee record and flagged, with both source records visible side-by-side.
- Each exception can be drilled into down to the source record in the HRIS app.
- Root cause classification and risk score are displayed per case.

- The case lifecycle (Flagged → Closed/Escalated) is fully visible and navigable for at least one completed example case.
- At no point does the Audit Monitoring Tool automatically deactivate a record, block a payment, or alter a payroll transaction.
- The normal/baseline path (Section 6) can be shown or narrated in under 30 seconds.

13. Open Questions / Assumptions for Review

- Exact activity-tolerance window to be used for ghost worker detection (e.g. days since last site-access/attendance log) — to be confirmed with the development and audit methodology leads.
- Scenario 9 was proposed by Deloitte as a supporting scenario outside NLNG's confirmed Contract Management / Asset & Maintenance Management scope and has not yet been validated directly with NLNG (Ikemefuna/Omowunmi) — recommend confirming before final build sign-off.
- Whether site-access or attendance log data is realistically available/simulatable for the ghost worker rule, or whether detection should rely solely on HR exit records versus payroll status.
- Whether the Audit Monitoring Tool's case management framework (Section 9) is being built centrally for all 10 scenarios or needs scenario-specific instancing — to be confirmed with the development lead.